EXECUTIVE PRESENTATION

CUSTOMER SEGMENTATION & CLV ANALYSIS

UK E-Commerce | 4,338 Customers | Dec 2010 – Dec 2011

4,338
Customers

£8.89M
Total Revenue

4
Segments

£139K
Champion CLV

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THE OPERATIONAL PROBLEM

WASTED BUDGET

The business runs identical campaigns for every customer — same offers, same frequency, same messaging — regardless of how much they spend or how often they buy.

Marketing waste · Margin erosion · No prioritisation

MISSED RETENTION

High-value customers — the minority driving the majority of revenue — receive no special treatment. They are one bad experience away from churning, with no safety net in place.

Churn risk · No loyalty signals · Revenue cliff

"Treating all customers identically is not neutral — it is a decision to over-invest in the wrong ones and under-invest in the ones that matter."

THE DATASET AT A GLANCE

541,909
Raw Transactions

392,692
Clean Transactions

4,338
Unique Customers

13 Months
Dec 2010 – Dec 2011

DATA CLEANING SUMMARY

135,080 missing CustomerIDs dropped (24.9%) · 9,288 cancelled invoices removed · 5,268 duplicate records removed · total_price engineered as (quantity × unit price)

RAW → CLEAN

541,909 transactions reduced to 392,692 after removing cancellations, missing IDs, negatives, and duplicates. 27.6% of raw data excluded — all for defensible, documented reasons.

CUSTOMER SCOPE

4,338 unique customers retained. Each customer aggregated into a single RFM profile — Recency, Frequency, and Monetary — the three dimensions that define behavioural value.

SNAPSHOT DATE

Set to 2011-12-10 (day after last transaction) as the reference point for recency calculation. Ensures consistency across all customer-level time comparisons.

WHAT THE DATA REVEALS: REVENUE STRUCTURE

REVENUE CONCENTRATION

74.66%

of total revenue driven by the top 20% of customers

The remaining 80% of customers contribute just 25.34% of revenue. Revenue is not evenly distributed — it is structurally concentrated in a small minority.

ONE-TIME BUYERS

34.42%

of customers placed only a single order — the largest group in the dataset

DORMANT CUSTOMERS (180d+)

860

customers inactive 180+ days, contributing only 6.24% of total revenue

"Revenue is highly concentrated — the top 13% of customers (Champions) alone carry 63% of the business."

THE FOUR SEGMENTS

K=4 | Silhouette 0.57

CHAMPIONS

571
customers

Revenue
63.12%

Avg Recency
20 days

Avg Frequency
15.8 orders

Avg Spend
£9,823 avg spend

13% of customers. 63% of revenue. The business.

LOYAL CUSTOMERS

1,452
customers

Revenue
26.86%

Avg Recency
46 days

Avg Frequency
4.2 orders

Avg Spend
£1,644 avg spend

Established habit. Gap to Champions is frequency, not trust.

POTENTIAL LOYALISTS

1,377
customers

Revenue
5.95%

Avg Recency
58 days

Avg Frequency
1.51 orders

Avg Spend
£384 avg spend

Recent but almost all one-time buyers. Still warm.

AT RISK

938
customers

Revenue
4.07%

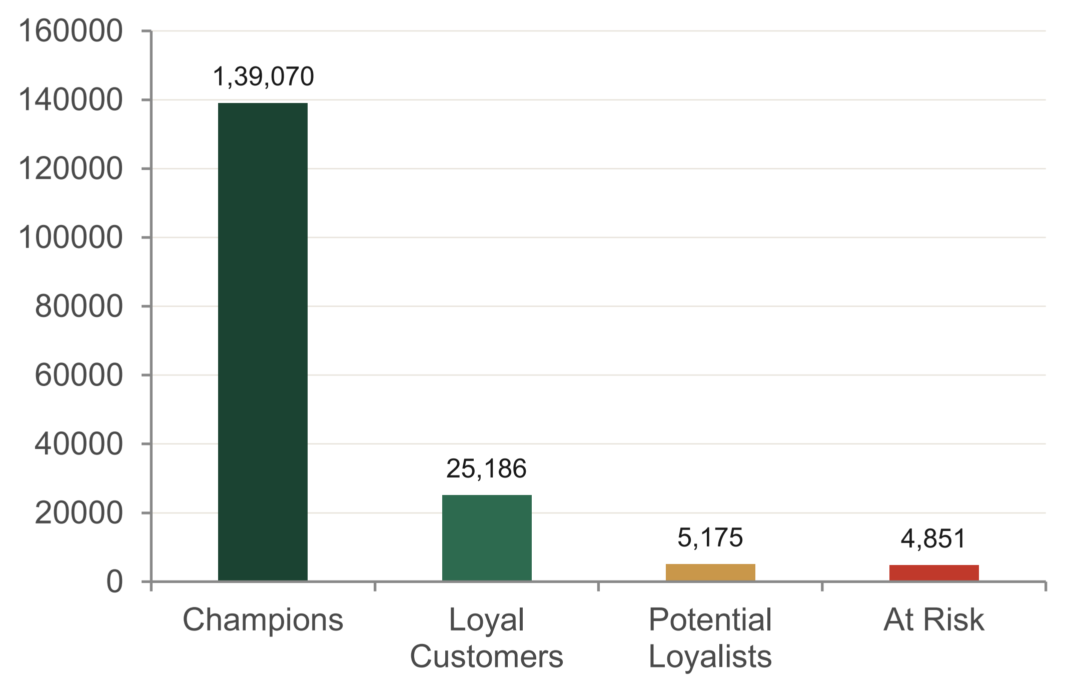
Avg Recency
259 days

Avg Frequency
1.37 orders

Avg Spend
£385 avg spend

Silent for 8+ months. Low value but recoverable.

THE CLV GAP IS EXTREME



£139,070
Champions CLV
Annualised proxy

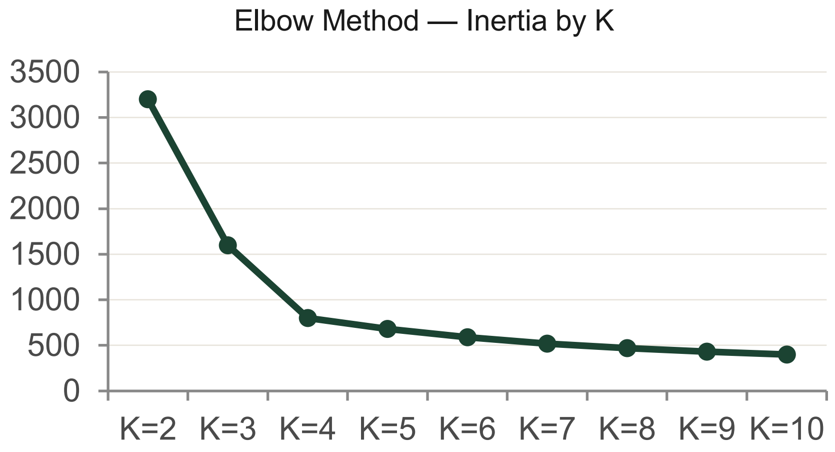
5.5×
vs Loyal Customers
CLV multiplier

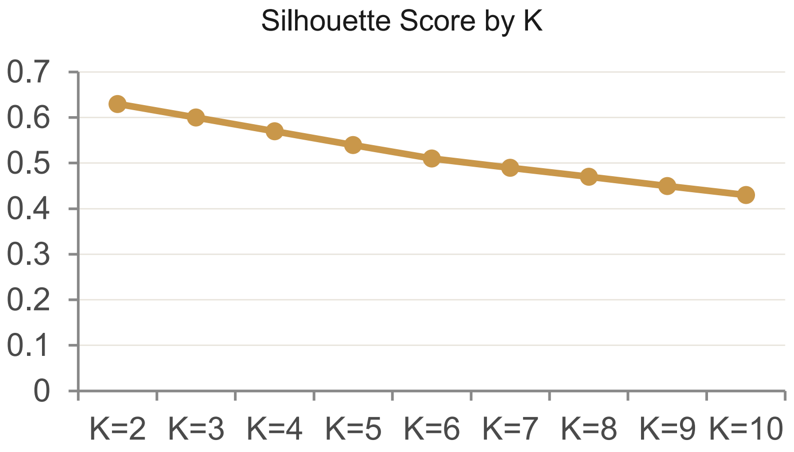
28×
vs At Risk
CLV multiplier

Low order frequency is the primary drag on Potential Loyalists — 1.51 orders vs 15.8 for Champions.

"Every ₹ of marketing budget allocated without CLV context is a ₹ misdirected."

WHY K=4 WAS SELECTED





ELBOW BEND AT K=4

Inertia drops sharply from K=2 (3.2B) to K=4 (0.8B) then flattens. The clearest bend in the curve sits between K=4 and K=5.

SILHOUETTE SCORE 0.57

K=2 mathematically peaks at 0.63 but collapses all nuance into 2 groups. K=4 holds a strong 0.57 — meaningful separation before the sharper drop.

BUSINESS INTERPRETABILITY

K=4 maps cleanly onto actionable business segments — Champions, Loyal, Potential Loyalists, At Risk — each requiring a distinct strategy.

K=4 selected on convergence of statistical signal and business interpretability — not on either criterion alone.

STRATEGIC RECOMMENDATIONS

Every recommendation is traceable to a specific number.

01

Protect the Champion Base

571 customers · 63.12% revenue

Build a VIP retention programme — early access, dedicated support, loyalty rewards tied to frequency. Any Champion crossing 45+ days without a purchase triggers immediate re-engagement.

02

Convert Loyals to Champions

1,452 customers · 4.2 → 15.8 order gap

Run frequency-focused campaigns — bundle offers, repeat-purchase incentives. Identify the top 20% of Loyal Customers as Champion candidates and give them early product access.

03

Activate Potential Loyalists

1,377 customers · 58-day recency window

Deploy a second-purchase campaign targeting these users within 30 days of first order. A small incentive at the right moment converts a browser into a repeat buyer.

04

Re-engage At Risk Customers

938 customers · 259 days inactive

Win-back sequence with a time-limited offer. If response rate is below 5%, reallocate budget to Potential Loyalists with higher conversion probability.

05

Never Discount Champions

Avg spend £9,823 · already paying full price

Champions are high-frequency, high-spend buyers. Blanket discounts erode margin with zero incremental value. Reserve discounts exclusively for activation and re-engagement.

06

Prioritise Frequency Over Spend

CLV = AOV × Frequency × 12

Frequency is the primary driver of segment value. Investment that increases order frequency, even at lower AOV, compounds into significantly higher lifetime value.

07

Allocate Budget by CLV

£139K vs £25K vs £5K vs £4.8K

A uniform per-customer budget treats a £139K customer identically to a £4.8K one. Marketing spend should reflect the 28× CLV gap — not be applied uniformly.

08

Monitor Revenue Concentration Risk

Top 20% = 74.66% of revenue

If Champions churn, no other segment compensates. Track Champion recency weekly and set board-level alerts at cohort-level churn signals.

"Every recommendation in this project has a number behind it. That is the difference between analysis and opinion."

"The segmentation doesn't tell you what to do. It tells you who your customers actually are — and what each group is worth. Strategy follows from that."

Revenue Is Concentrated

Top 20% drive 74.66% of revenue. Champions alone carry 63%. The imbalance is structural and measurable.

Segments Are Distinct

Four behaviorally different groups with a 28× CLV gap end-to-end. Same model, four different strategies.

The Actions Are Clear

Every recommendation is grounded in a specific number — CLV, recency, frequency. No guesswork.